# Question ID: 35b46381

| Assessment | Test | Domain | Skill | Difficulty |
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| SAT | Reading and Writing | Information and Ideas | Central Ideas and Details | Hard |

**Question**

In their 2022 paper, Christos Dimopoulos et al., having granted that the existence of antigravity—in which antimatter and matter repel rather than attract each other—lacked affirmative experimental support, rightly argued that such antigravity was worth considering on theoretical grounds given that evidence against it was similarly lacking. But a 2023 report by an international team of researchers details the first direct ballistic observations of antihydrogen atoms under gravity inside a CERN particle accelerator. Corresponding most closely to predictions under gravitational attraction, these observations were thoroughly inconsistent with antigravity.

Which choice best states the main idea of the text?

**Answer**

A. Antihydrogen ballistics observations were conducted at CERN to test specific conclusions about antigravity presented in the 2022 paper by Dimopoulos et al.

B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report's findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.

C. The theoretical approach represented in the 2022 paper by Dimopoulos et al. assumed that unambiguous proof of antigravity would not be achievable, but the results in the 2023 report undermine that assumption.

D. Before 2023, researchers' inordinate focus on theoretical considerations hindered the development of the experimental regimen for direct antihydrogen ballistics observations.

**Correct Answer: B**

**Rationale**

Choice B is the best answer because it most accurately states the main idea of the text. The text begins by acknowledging that in 2022, Dimopoulos et al. were justified in arguing for antigravity on theoretical grounds because although there was no experimental evidence supporting its existence at that time, there was no evidence contradicting its existence either. The text then introduces the 2023 report describing an experiment that produced observations "thoroughly inconsistent with antigravity." Thus, the main idea is that while the study of antigravity was theoretically justified before the 2023 report (due to lack of evidence against it), the report's findings undermine the rationale for such theoretical consideration since there is now evidence against this hypothetical phenomenon.

Choice A is incorrect. Although antihydrogen ballistics observations were conducted at CERN, the text provides no evidence that they were conducted to test conclusions in Dimopoulos et al.'s 2022 paper. Rather, the text presents the 2023 report as new evidence that contradicts the existence of the hypothetical phenomenon that Dimopoulos et al. had considered. Choice C is incorrect because the text doesn't suggest that Dimopoulos et al. assumed that unambiguous proof of antigravity would be unattainable. Rather, the text states that they argued antigravity was worth considering theoretically because evidence against it was lacking at that time. At the same time, the results in the 2023 report do not undermine but strengthen the idea that the proof of antigravity might not be achievable. Choice D is incorrect because the text doesn't suggest that there was an "inordinate focus on theoretical considerations" before 2023 or that such a focus hindered the development of experimental methods. In fact, the text describes Dimopoulos et al.'s position regarding the theoretical consideration of antigravity as "rightly argued," indicating approval of their approach given the absence of experimental evidence at that time.